

Developing a Culturally Adapted Adolescent Life-Skills Curriculum (THRIVE-Belize) Using Community-Based Participatory Research and AI-Assisted Synthesis

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Main message: AI agents read 38 public manuals, more than 5,000 pages, so a small team could choose the best activities for one Belizean school.

1. The need

Across the English-speaking Caribbean, about one in four adolescents (23.5%) has a mental-health problem, according to a meta-analysis of 28 studies with 65,034 adolescents.¹ A review of 50 randomized trials in 19 low- and middle-income countries found that life-skills programs improved life skills and functioning and reduced depression and anxiety.²

In 2025, Hillside Health Care surveyed students at Toledo Community College. Risks clustered within the same students, and each of the seven planned modules matched a need the survey measured. These results are not yet published.

Our question. Can AI agents help researchers find, read, and choose activities from long public manuals to build a curriculum for one school in the Toledo District?

2. Who did what

Mario Morales (MM) designed the AI workflow and wrote its rules, set the inclusion criteria, and, with Aimee, approved each module. Aimee Slagle (AS) checked random samples against the original manuals and advised on cultural adaptation. Gabriela Valdez (GV), co-principal investigator, provided institutional support. The AI agents worked in three roles. The Researcher extracted every activity. The Architect checked each extraction against the PDF. The Local Expert flagged problems with culture, safety, and cost.

3. From pages to sessions, January to February 2026

Figure 1 (text version). How the evidence narrowed

Stage	Count
Records from an AI web search of 27 queries (17 January 2026)	60
Pages in the 38 free manuals the agents read in full	5,074
Activities catalogued, each with a quote and a page number	More than 1,100
Activities chosen for the THRIVE-Belize manual	95

Figure 2 (text version). The agents worked in four steps.

1. **Find (MM and AI).** In January 2026, the agent searched for free manuals for grades 6 to 12 that held real classroom activities. The criteria also required English, school use, and publication between 2015 and 2026.
2. **Read (MM and AI).** Following Mario's rules, the agents cut each PDF into 10-page pieces and recorded every activity in 17 fields, including a verbatim quote and the page range.
3. **Check (AS).** In the first three manuals the agents processed, Aimee compared a random 20% of the activities with the original manuals.
4. **Choose (AI, MM, and AS).** The agents ranked activities by fit to the school, and Mario and Aimee approved each module before the next one began.

4. The result

Figure 3 (text version). The seven modules

Module	Sessions
M01 Communication and emotions	2-5
M02 Gender norms and boys' health	6-9
M03 Sexual and reproductive health	10-13
M04 Healthy relationships	14-17
M05 Mental and physical health	18-21
M06 Substance use and refusal	22-25
M07 Environment and climate	26-29

Session 1 opens the program, session 30 closes it, and a booster session follows 8 to 12 weeks later. That makes 31 sessions of 40 minutes, all paper-based and low-cost.

5. Adaptation

We rewrote activities for Toledo and logged every change. Three examples follow, with the entry number in the manual's adaptation log.

- A generic food chain became sardine, snapper, and barracuda, fish that students know and eat (entry 69).
- A historical case study now includes designing warning labels in English, Kriol, Spanish, Q'eqchi', Mopan Maya, and Garifuna (entry 70).
- Video scenarios became read-aloud scenarios, so no projector is needed (entry 30).

Rules the AI could not override included the right to pass, never promising secrecy, fictional scenarios only, a quote and page for every activity, and exactly four sessions per module.

6. Where we are now

We completed Phase 1 in July and August 2026. Students, teachers, parents, school leaders, and community members gave feedback in surveys, focus groups, and interviews, and we presented the manual to key stakeholders.

Take-home: with clear criteria and careful checks, AI agents let a small team read thousands of pages and build a curriculum in which every activity traces to its source.

We learned two lessons. Aimee's spot checks found activities that the agents had skipped or partly invented. We then required them to extract every activity and to show that three counts matched. A final read-through found timing errors that the automated checks had passed, and we logged them for correction.

References

1. Liverpool S, Prescod J, Pereira B, Trotman C. Prevalence of mental health and behaviour problems among adolescents in the English-speaking Caribbean: systematic review and meta-analysis. *Discov Ment Health*. 2023;3(1):11. doi:10.1007/s44192-023-00037-2
2. Singla DR, Waqas A, Hamdani SU, Suleman N, Zafar SW, Zill-E-Huma, et al. Implementation and effectiveness of adolescent life skills programs in low- and middle-income countries: a critical review and meta-analysis. *Behav Res Ther*. 2020;130:103402. doi:10.1016/j.brat.2019.04.010

AI disclosure

This work studies the use of artificial intelligence. AI agents searched the web for manuals, extracted their activities, and drafted manual content under our written rules in January and February 2026. We checked their work against the original

manuals and approved every module. AI also helped draft this poster and handout, and we verified all of the content.

Declarations

Ethics. Phase 0 (2025 survey) was approved by the Belize Ministry of Health and Wellness IRB (GEN/147/01/25(42)Vol.VI). Phase 1 (2026) was approved by the University of Arizona IRB (STUDY00007955) and the Belize Ministry of Health and Wellness IRB (MOHW/IRB/2026/002), and authorized by the Belize Ministry of Education, Culture, Science and Technology (GEN/21/26(112) VOL. III). Phase 1 is registered at ClinicalTrials.gov (NCT07687199). **Funding.** University of Arizona Graduate and Professional Student Council (GPSC) Spring 2026 Graduate College ReaP Grant. **Conflicts of interest.** None declared. **Contact.** mariomorales@arizona.edu.